1. **Write a C program to insert an element from the any position of an array.**

```
#include<stdio.h>
int main(void){
    int n,pos,element;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements into the array: \n");
    for(int i=0;i<n;i++){
        scanf("%d", &arr[i]);
    }
    printf("Entered array: \n");
    for(int i=0;i<n;i++){
        printf("%d ", arr[i]);
    }
    printf("\nEnter the element to be inserted: ");
    scanf("%d", &element);
    printf("Enter the position: ");
    scanf("%d", &pos);
    if(pos<1 || pos>(n+1)){
        printf("Invali Position");
    }else{
        for(int i=n;i>=pos;i--){
            arr[i]=arr[i-1];
        }
    }
    arr[pos-1] = element;
    n++;
    printf("Updated array: \n");
    for(int i=0;i<n;i++){
        printf("%d ", arr[i]);
    }
    return 0;
}
```

2. **Write a C program to delete an element from the any position of an array.**

```
#include<stdio.h>
int main(void){
    int n,pos;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements into the array: \n");
    for(int i=0;i<n;i++){
        scanf("%d", &arr[i]);
    }
    printf("Entered array: \n");
    for(int i=0;i<n;i++){
        printf("%d ", arr[i]);
    }
    printf("\nEnter the position: ");
    scanf("%d", &pos);
```

```
    for(int i=pos-1;i<=(n-1);i++){
        arr[i]=arr[i+1];
    }
    n--;
    printf("Updated array: \n");
    for(int i=0;i<n;i++){
        printf("%d ", arr[i]);
    }
    return 0;
}
```

3. **Write a C program to swap first and last digits of a number.**

```
#include <stdio.h>
#include <math.h>

int main() {
    int num, n, firstDigit, lastDigit, digits = 0, power,
middlePart;

    printf("Enter a number: ");
    scanf("%d", &num);

    n = num;

    // Find number of digits
    while (n > 0) {
        n /= 10;
        digits++;
    }

    // If number has only 1 digit, no need to swap
    if (digits == 1) {
        printf("After swapping: %d\n", num);
        return 0;
    }

    // Get last digit
    lastDigit = num % 10;

    // Get first digit
    firstDigit = num;
    while (firstDigit >= 10) {
        firstDigit /= 10;
    }

    // Remove first and last digits from number
    power = pow(10, digits - 1);
    middlePart = (num % power) / 10;

    // Create the new number with swapped digits
    int swapped = lastDigit * power + middlePart * 10 + firstDigit;

    printf("After swapping: %d\n", swapped);
```

```
    return 0;
}
```

4. **Transpose of a matrix.**

```
//Transpose of a matrix

#include<stdio.h>
int main(void){
    int m,n;
    printf("Enter the number of rows and columns: ");
    scanf("%d%d", &m,&n);
    int arr[m][n], tra[n][m];
    printf("Enter elements into the matrix: ");
    for(int i=0;i<m;i++){
        for(int j=0;j<n;j++){
            scanf("%d", &arr[i][j]);
        }
    }
    printf("Entered matrix is: \n");
    for(int i=0;i<m;i++){
        for(int j=0;j<n;j++){
            printf("%d\t", arr[i][j]);

        }
        printf("\n");
    }
    for(int i=0;i<m;i++){
        for(int j=0;j<n;j++){
            tra[j][i] = arr[i][j];
        }
    }
    printf("Transpose of the matrix is: \n");
    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            printf("%d\t", tra[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

5. **Write a *C* program to count total number of vowels and consonants in a string.**

```
#include<stdio.h>
#include<string.h>
int main(void){
    char str[100], str2;
    int vowels = 0, consonents = 0;
    printf("Enter the string: ");
    scanf("%[^\n]", str);
    for(int i=0;str[i] != 0;i++){
```

```
        str2 = tolower(str[i]);
        if(str2 >= 'a' && str2 <= 'z'){
            if(str2 == 'a' || str2 == 'e' || str2 == 'i' || str2 ==
'o' || str2 == 'u'){
                vowels++;
            }else{
                consonents++;
            }
        }
    }
    printf("Number of\nVowels: %d\nConsonents: %d\n", vowels,
consonents);
    return 0;
}
```

6. **Write a C program to find reverse of a string.**

```
#include<stdio.h>
#include<string.h>
int main(void){
    char str1[100];
    printf("Enter the string: ");
    scanf("%[^\n]", str1);
    int length = strlen(str1);
    for(int i=0;i<length /2;i++){
        char temp = str1[i];
        str1[i] = str1[length - i- 1];
        str1[length - i -1] = temp;
    }
    printf("Resultant string is: %s", str1);
    return 0;
}
```

7. **Write a C program to check whether a string is palindrome or not (using single string only)**

```
#include<stdio.h>
#include<string.h>
int main(void){
    char str1[100], copy[100];
    printf("Enter the string: ");
    scanf("%[^\n]", str1);
    strcpy(copy, str1);
    int length = strlen(str1);
    for(int i=0;i<length /2;i++){
        char temp = str1[i];
        str1[i] = str1[length - i- 1];
        str1[length - i -1] = temp;
    }
    if(strcmp(str1, copy) == 0){
        printf("It is a Palindrome");
    }else{
        printf("It is not a Palindrome");
```

```
    }
    return 0;
}
```

8. **Write a C Program to count the frequency of array elements in a 1-D array**

```
#include<stdio.h>
int main(void){
    int n, target,count =0;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements into the array: \n");
    for(int i=0;i<n;i++){
        scanf("%d", &arr[i]);
    }
    printf("Enter the target number: ");
    scanf("%d", &target);
    for(int i=0;i<n;i++){
        if(arr[i] == target){
            count++;
        }
    }
    printf("The frequency of the target element is: %d", count);
    return 0;
}
```

9. **Write a C program to find the Fibonacci series up to n term using recursive function**

```
#include<stdio.h>
int fibo(int n){
    if(n==0){
        return 0;
    }else if(n==1){
        return 1;
    }else{
        return fibo(n-1) + fibo(n-2);
    }
}
int main(void){
    int n;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        printf("%d ", fibo(i));
    }
    return 0;
}
```